

1. String Rotation A *rotation* of a string α is the string $\alpha[k :]\alpha[: k]$ (the last $|\alpha| - k$ characters of α preceding its last k) for some index k .

- a. Propose a linear-time algorithm which, given two strings α and β , decides whether α is a rotation of β for some k .
- b. It should be relatively clear that a **True** response implies that α is a rotation of β . Prove the following:
Proposition. *If my process returns **False**, then α is not a rotation of β .*
- c. Why does your process fail when looking for a rotation of β *contained in* α ? Give an example.